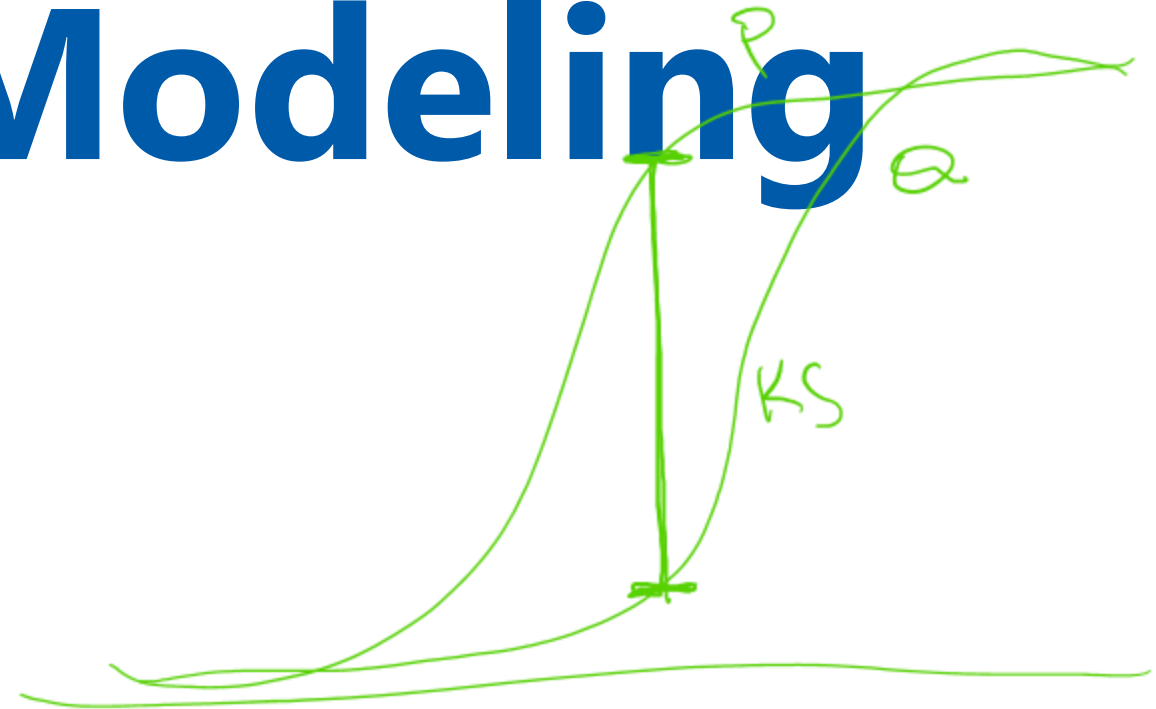


Generative Modeling

Distances and metrics

Tigran Ramazyan, Anwar Ibrahim, Fedor Pakhurov

Laboratory for methods of big data analysis



In this Lecture

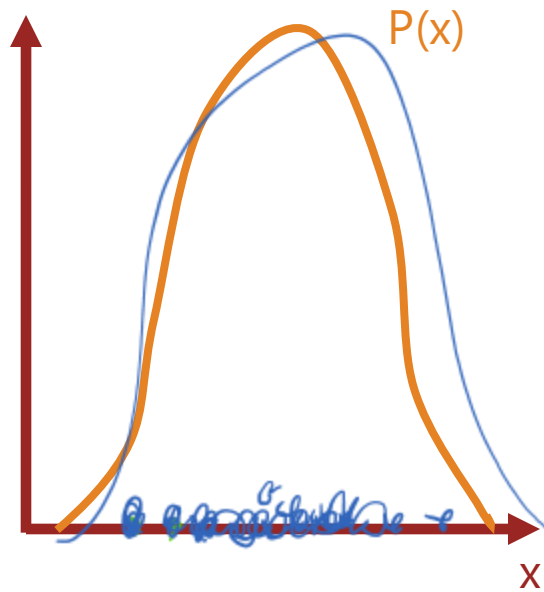
- ▶ f -divergences
 - total Variation Distance;
 - Kullback-Leibler Divergence;
 - Jensen-Shannon Divergence;
 - divergence inequalities;
 - variational lower bound.
- ▶ Metrics (seminar)

Total Variation Distance

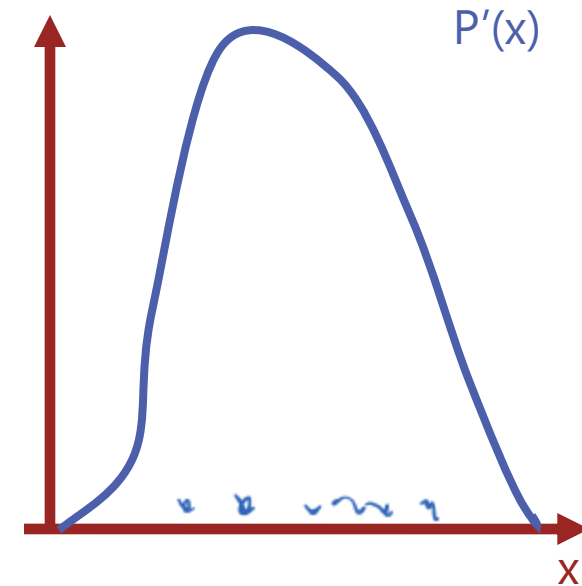


What we measure

MSE
BCE $\|y - \hat{y}\|$



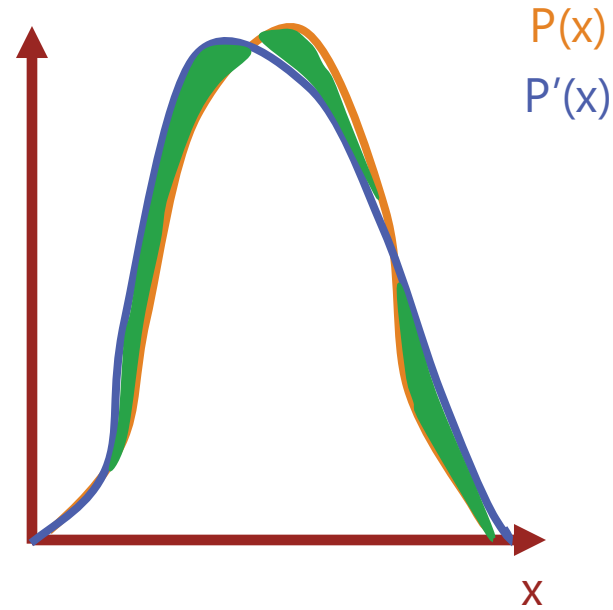
True Probability Density



Fitted Probability Density

$P'(x)$ is similar to $P(x)$?

First idea: absolute difference



$$\int_A^y |P(x) - \hat{P}(x)| dx = 1.$$

Total Variation Distance^{TV}

For $p(x)$ and $q_\theta(x)$ being PDFs:

$$D(p(x), q_\theta(x)) = \boxed{\frac{1}{2}} \int |p(x) - q_\theta(x)| dx$$

This can be rewritten using Scheffe's theorem

$$D(p(x), q_\theta(x)) = \sup_A \left| \int_A p(x) dx - \int_A q_\theta(x) dx \right|$$

$$\int p = 1$$

$$\int q = 0$$

$$\int p = 0$$

$$\int q = 1$$

Where A is any measurable set.

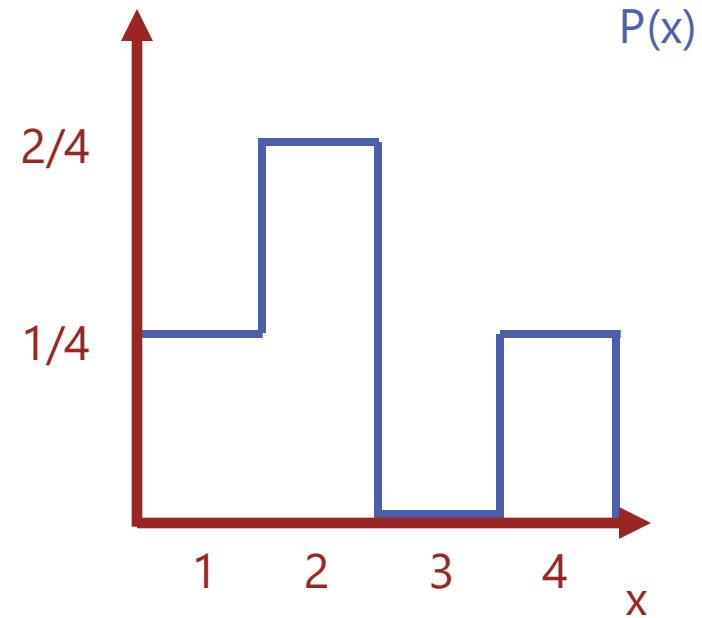
$$A = B \cup C$$

A. B. Tsybakov, Introduction to Nonparametric Estimation, sec 2.4

$$p = q \Leftrightarrow TV(p, q) = 0$$

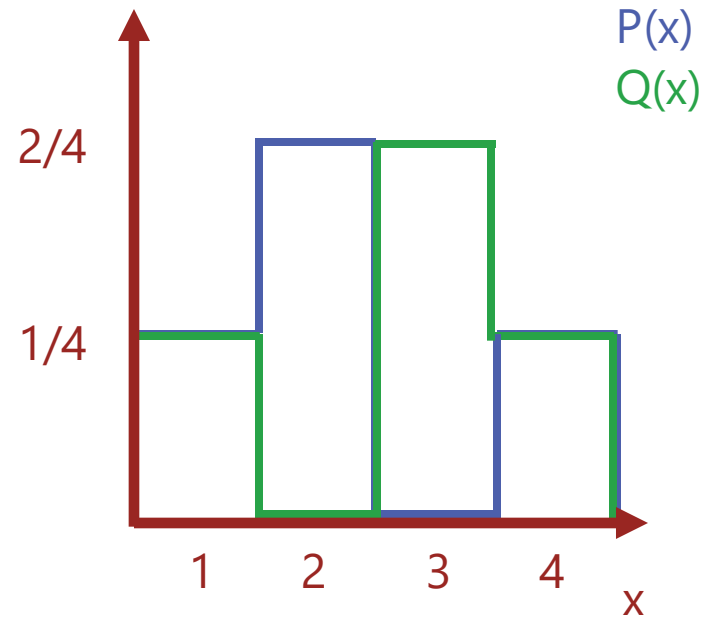
Total Variation Distance: example 1D

- discrete case for two PDFs



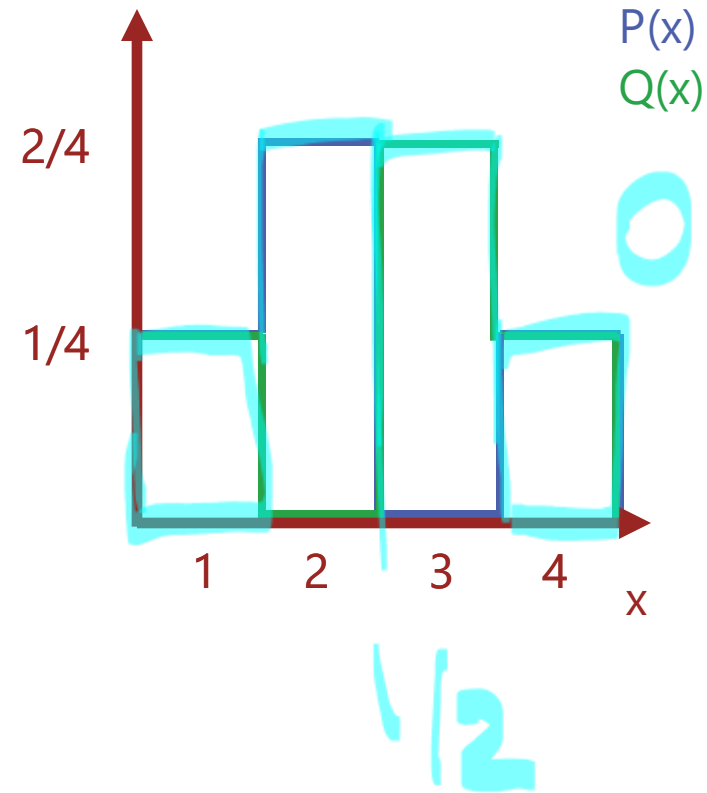
Total Variation Distance: example 1D

- discrete case for two PDFs



Total Variation Distance: example 1D

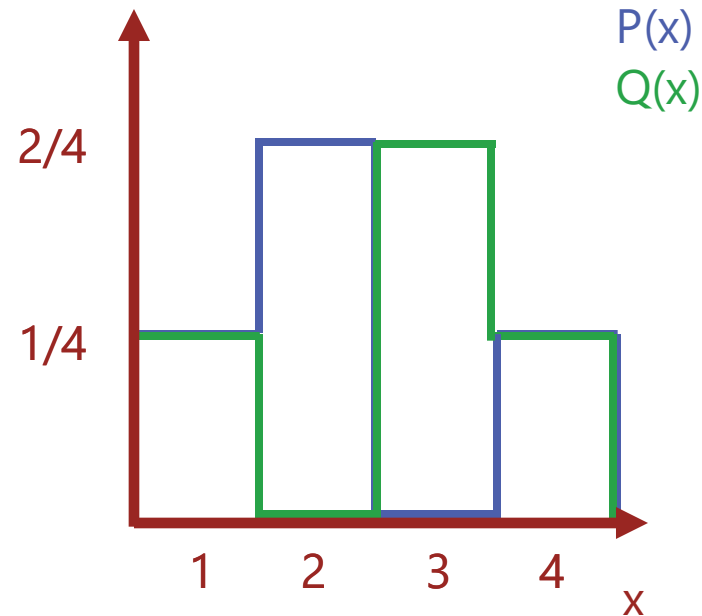
- discrete case for two PDFs
- calculate in two ways:



Total Variation Distance: example 1D

- discrete case for two PDFs
- calculate in two ways:
 - construct all possible subsets:

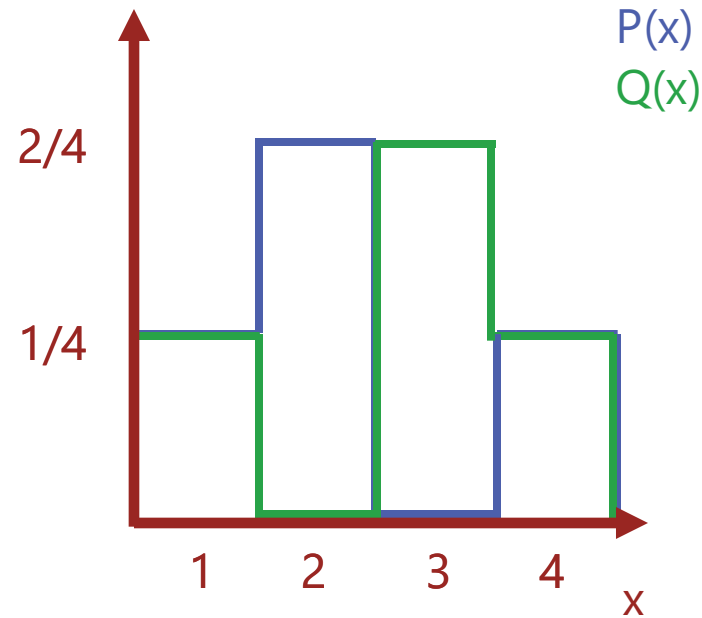
$\{1\}, \{2\}, \{3\}, \{4\}, \{1;2\}, \{1;3\}, \{1;4\},$
 $\{2;3\}, \{2;4\}, \{3;4\}, \{1;2;3\}, \{1;2;4\},$
 $\{1;3;4\}, \{1,2,3,4\}.$



Total Variation Distance: example 1D

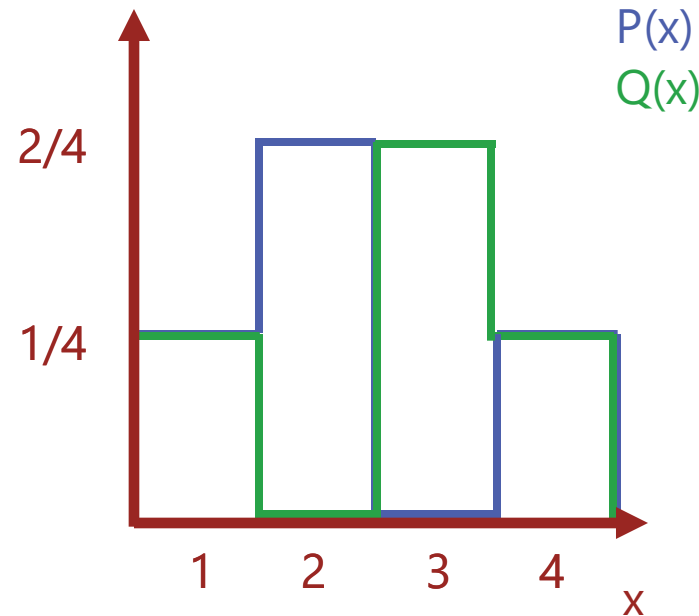
- discrete case for two PDFs
- calculate in two ways:
 - construct all possible subsets:

$$D(p,q) = 0.5$$



Total Variation Distance: example 1D

- discrete case for two PDFs
- calculate in two ways:
 - construct all possible subsets:
$$D(p,q) = 0.5$$
 - integrate over full range:
$$D(p,q) = 0.5$$



Total Variation Distance: observations

- Symmetric $D(p, q) = D(q, p)$
- Interpretable (using Scheffe lemma)
- Connected to hypothesis testing (D is the sum of errors)

$$\int |p - q| dx$$

Total Variation Distance: observations

- Symmetric $D(p, q) = D(q, p)$
- Interpretable (using Scheffe's theorem)
- Connected to hypothesis testing (D is the sum of errors)
- Too strong:

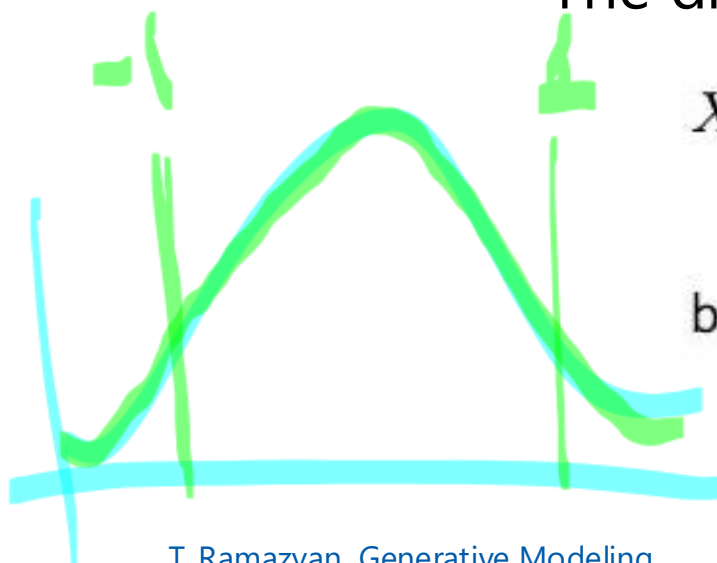
The distance might ignore the growing number of trials.

$X_1, \dots, X_n \sim \pm 1, S_n = \sum_n X_i$. Then

$$S_n / \sqrt{n} \rightarrow \mathcal{N}(0, 1),$$

but $D(S_n, \mathcal{N}(0, 1)) = 1$ for any n .

$\mathcal{N}(0, (\sqrt{n})^2)$

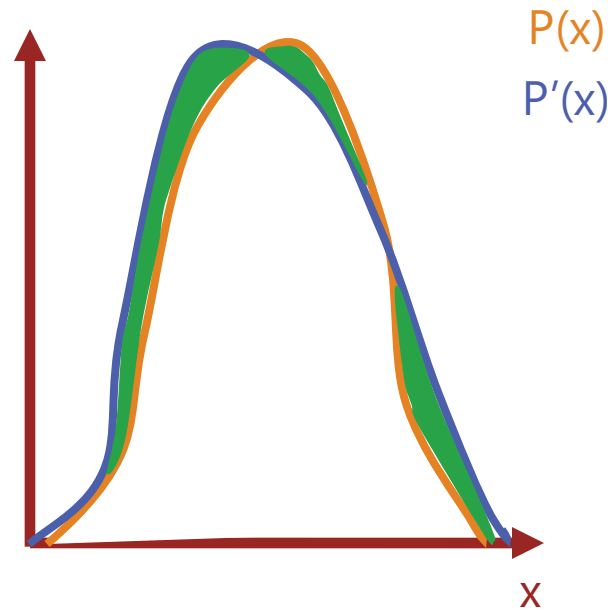


A. L. Gibbs, F. E. Su On Choosing and Bounding Probability Metrics
F Pollard, Total variation distance between measures

Kullback-Leibler Divergence



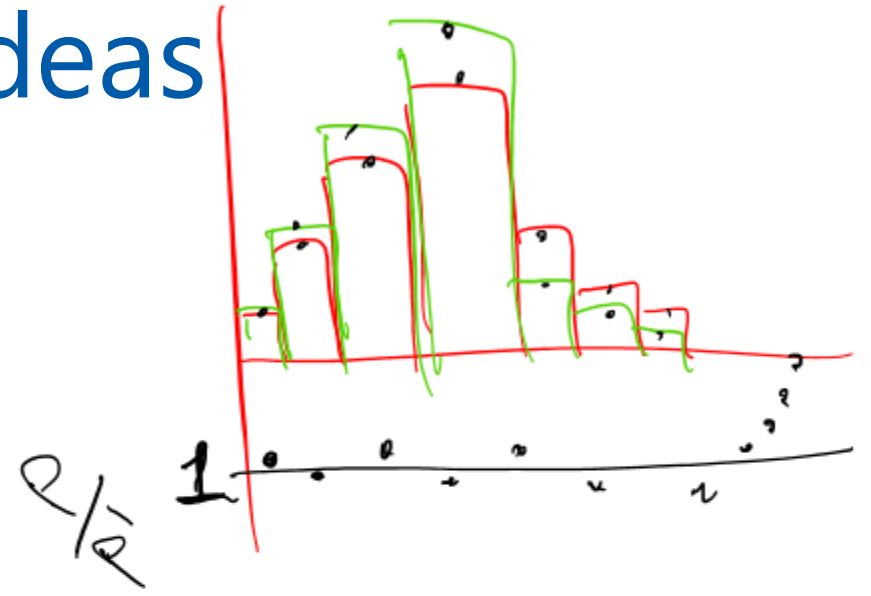
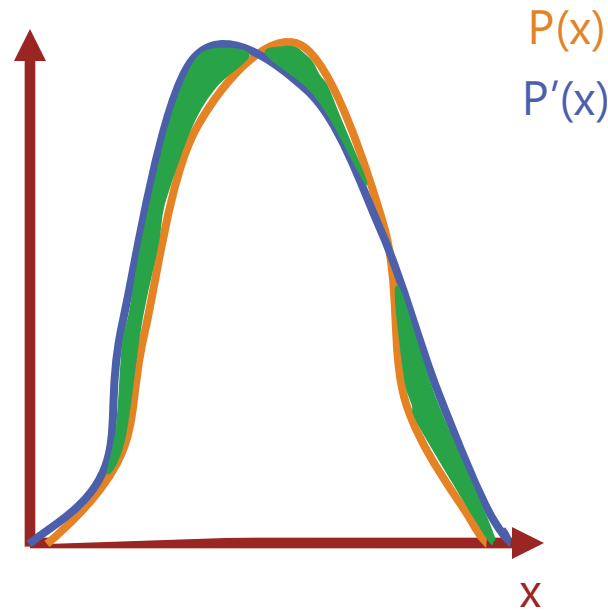
Kullback-Leibler divergence: ideas



Previously:

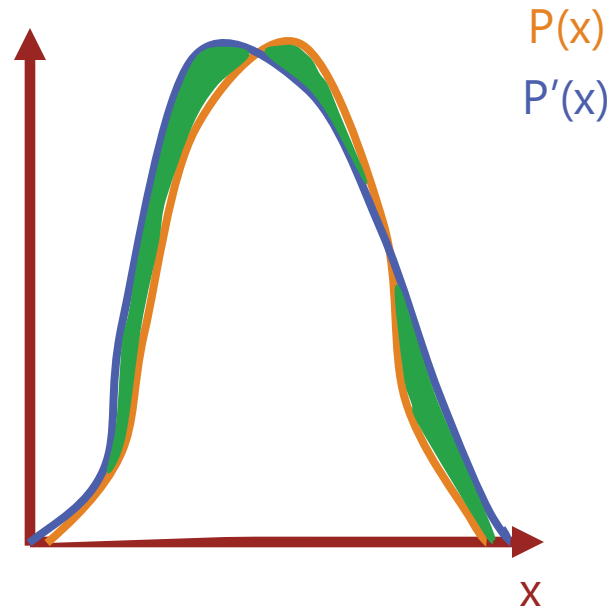
$$\int |P(x) - P'(x)| dx$$

Kullback-Leibler divergence: ideas



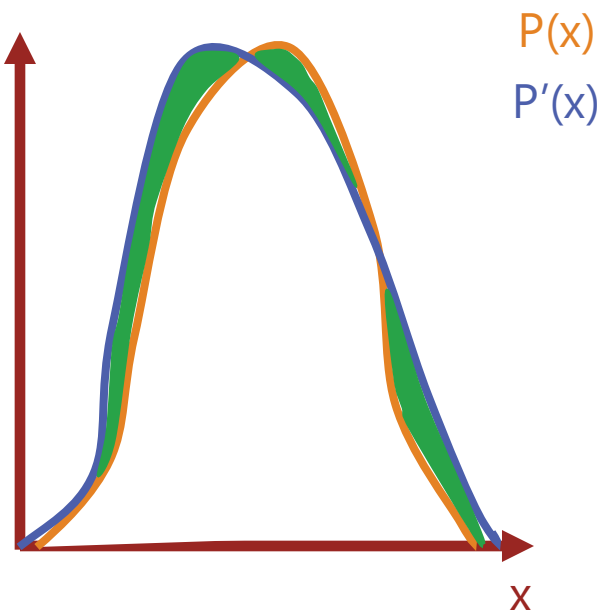
$$\ln \frac{P(x)}{P'(x)} = \ln P - \ln P'$$

Kullback-Leibler divergence: ideas



$$\ln \frac{P(x)}{P'(x)}$$

Kullback-Leibler divergence: ideas



$$\mathbb{E}[X|Y]$$

$$X \sim P$$

$$X' \sim P'$$

$$\int P(x) \ln \frac{P(x)}{P'(x)} dx$$

$$= \mathbb{E} \left[\ln \frac{P(x)}{P'(x)} \middle| X \right]$$

$x \sim P$

Kullback-Leibler divergence: definition

For $p(x)$ and $q(x)$, two probability distributions,

$$KL(p||q_\theta) = \int p(x) \log \left(\frac{p(x)}{q_\theta(x)} \right) dx$$

Kullback-Leibler divergence: definition

For $p(x)$ and $q(x)$, two probability distributions,

$$KL(p||q_\theta) = \int p(x) \log \left(\frac{p(x)}{q_\theta(x)} \right) dx$$

$$KL(Q||P) = \int q(x) \log (q(x)/p(x)) dx$$

- not symmetric $KL(P||Q) \neq KL(Q||P)$
- invariant under change of variables
- additive for independent variables
- nonnegative

$\forall A, B, C$

$$TV(A, B) \leq TV(A, C) + TV(C, B)$$

1. ≥ 0 + $= 0$ iff $P = Q$

~~2. symmetric~~

~~3. Her-бс тpeчeн.~~

Kullback-Leibler divergence: observations

- **KL divergence is connected to cross-entropy:**

$$KL(p||q) = \boxed{H(p)} + H(p, q),$$

where $H(p, q) = \mathbb{E}_p(\log q)$.

$$\begin{aligned} \int p(x) \log \frac{p(x)}{q(x)} dx &= \left[\int p(x) \log p(x) dx - \int p(x) \log q(x) dx \right] \\ &= \mathbb{E}_p[\log p(x)] + \mathbb{E}_p[\log q(x)] \end{aligned}$$

KL and Maximum Likelihood

Find the optimal parameter, θ^* :

$$\theta^* = \operatorname{argmin}_{\theta} KL(p(x) || q_{\theta}(x))$$

KL and Maximum Likelihood

Find the optimal parameter, θ^* :

$$\theta^* = \underset{\theta}{\operatorname{argmin}} KL(p(x) || q_{\theta}(x))$$
$$= \underset{\theta}{\operatorname{argmin}} (\mathbb{E}_{x \sim p} [\log p(x)] - \mathbb{E}_{x \sim p} [\log q_{\theta}(x)])$$

KL and Maximum Likelihood

Find the optimal parameter, θ^* :

$$\theta^* = \operatorname{argmin}_{\theta} KL(p(x) || q_{\theta}(x))$$

$$= \operatorname{argmin}_{\theta} (\underbrace{\mathbb{E}_{x \sim p}[\log p(x)] - \mathbb{E}_{x \sim p}[\log q_{\theta}(x)]})$$

$$= -\operatorname{argmin}_{\theta} \mathbb{E}_{x \sim p}[\log q_{\theta}(x)]$$

$$\operatorname{argmax}_{\theta} H(p, q_{\theta})$$

KL divergence: observations



- **KL divergence is connected to cross-entropy:**

$$KL(p||q) = H(p) + H(p, q),$$

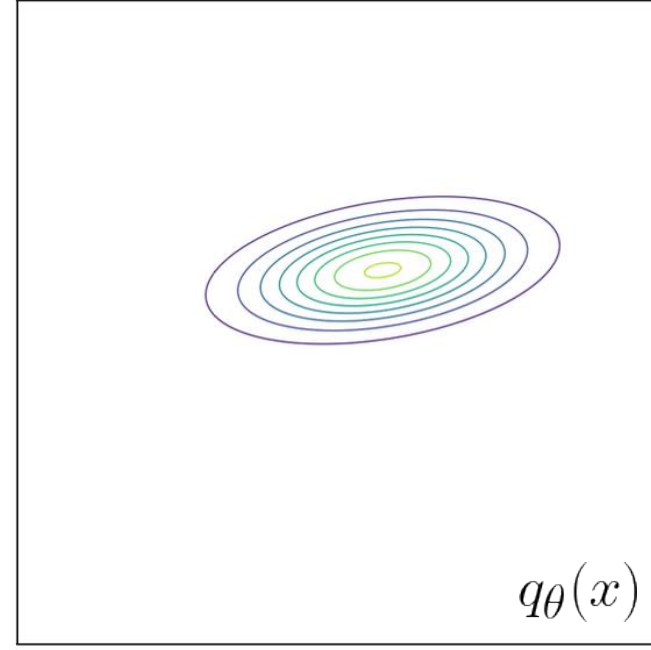
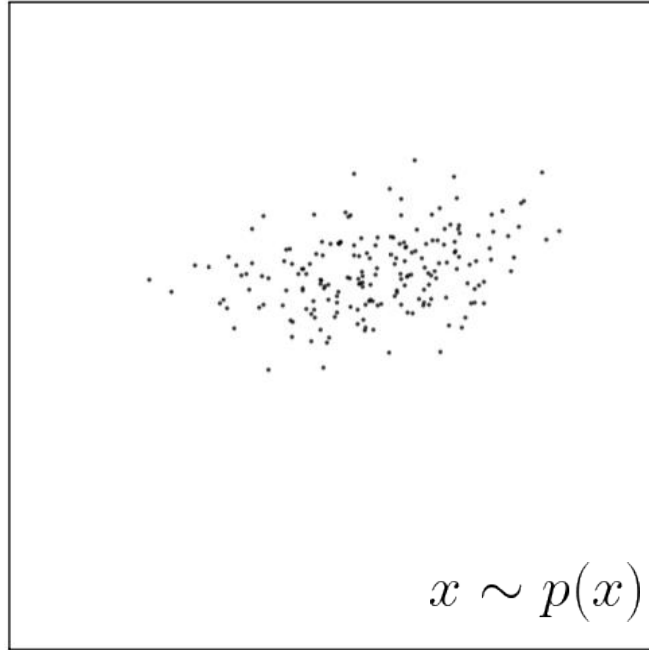
where $H(p, q) = \mathbb{E}_p(\log q)$.

- **Minimizing KL divergence is equivalent to maximizing the likelihood.**

$$\theta^* = \underset{\theta}{\operatorname{argmin}} KL(p(x)||q_{\theta}(x)) = \underset{\theta}{\operatorname{argmax}} \mathcal{L}(q_{\theta}(x); x)$$

Using in fits

Fit data points from 2D Gaussian function



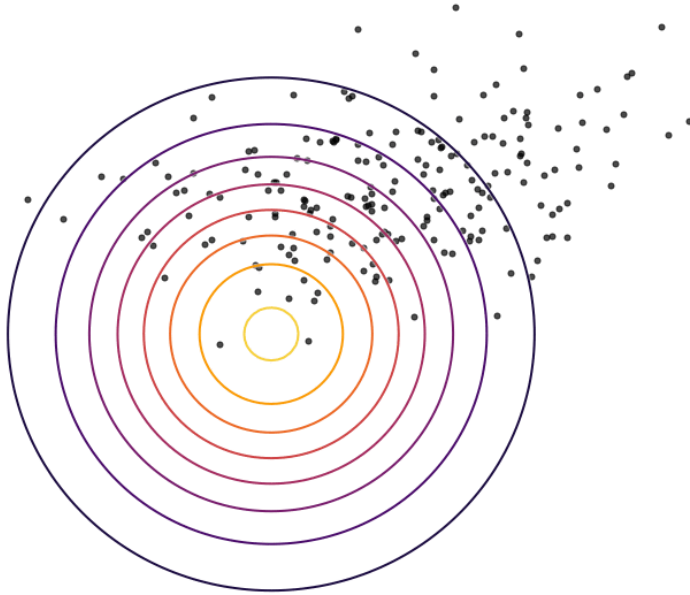
$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$$
$$\Sigma \in \mathbb{R}^{2 \times 2}$$

...with 2D Gaussian function

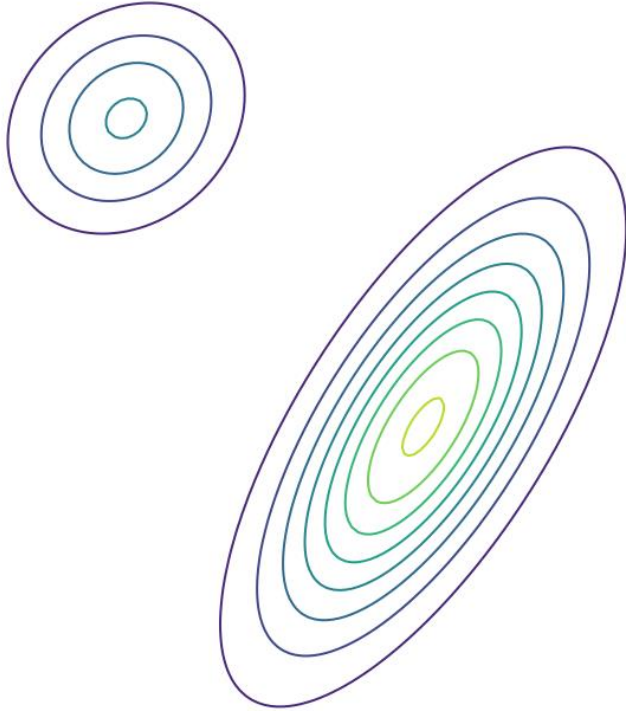
[Here and Later: Colin Raffel's blog](#)

Using in fits

- Runs smoothly for simple data

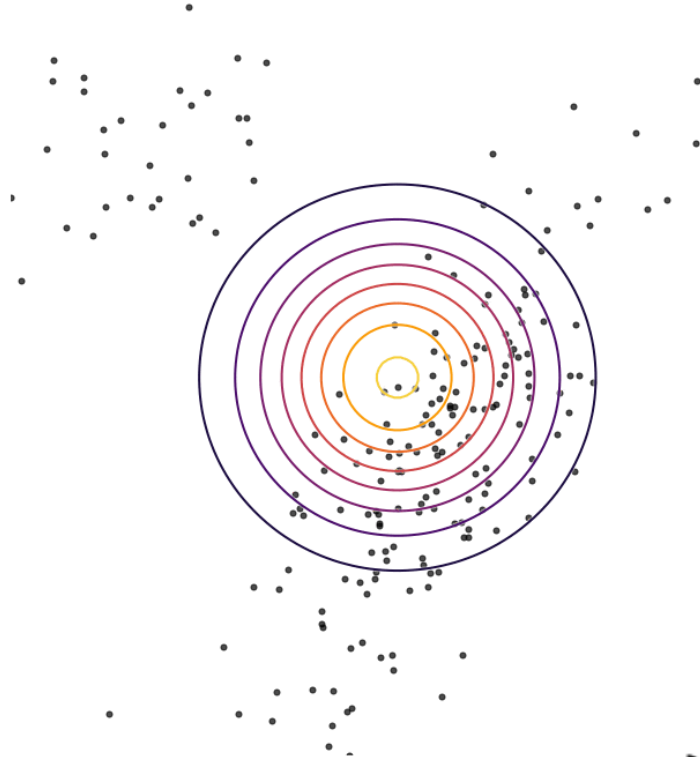


Using in fits: Multimodal data



- Runs smoothly for simple data

Using in fits: Multimodal data

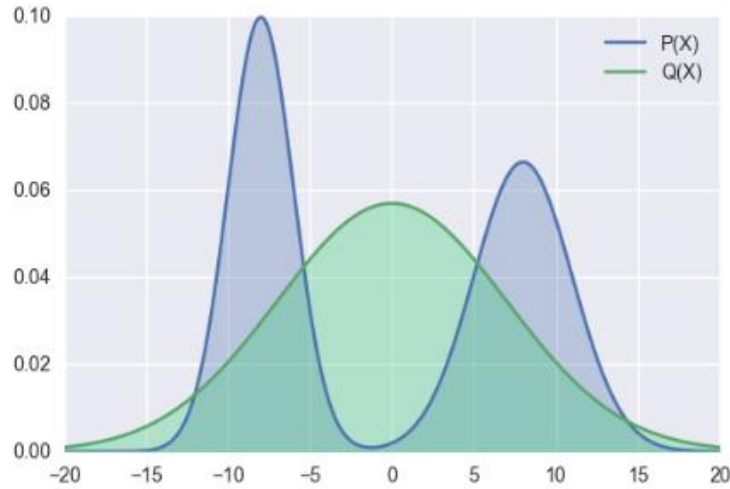


- Runs smoothly for simple data
- Problems for multimodal data
- Covers significant amount of empty spaces

$$\log \frac{p(x)}{q_{\theta}(x)}$$

KL divergence: study

$$KL(p||q_\theta) = \int p(x) \log \left(\frac{p(x)}{q_\theta(x)} \right) dx$$



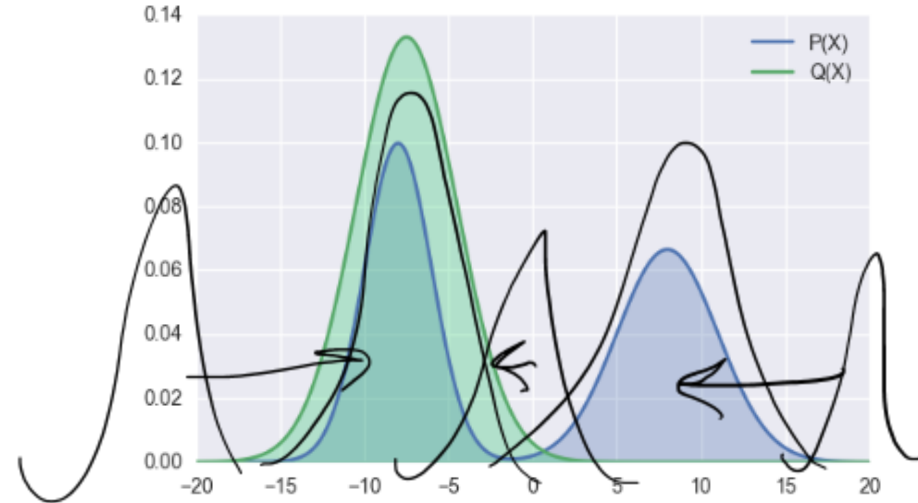
KL is zero avoiding, as it is avoiding $q(x) = 0$ whenever $p(x) > 0$

$$\log \frac{10^{-3}}{10^{-300}}$$

$$\ln \hat{p}(x)$$

$$\alpha N_1 + (1-\alpha) N_2$$

$$KL(q_\theta||p) = \int q_\theta(x) \log \left(\frac{q_\theta(x)}{p(x)} \right) dx$$



Reverse KL is zero forcing, as it forces $q(X)$ to be 0 on some areas, even if $p(X) > 0$

Reverse KL divergence: fits

Find the optimal parameter, θ^* :

$$\theta^* = \operatorname{argmin}_{\theta} KL(q_{\theta}(x) || p(x))$$

Reverse KL divergence: fits

Find the optimal parameter, θ^* :

$$\theta^* = \operatorname{argmin}_{\theta} KL(q_{\theta}(x) || p(x))$$

$$= \operatorname{argmin}_{\theta} (\mathbb{E}_{\tilde{x} \sim q_{\theta}} [\log q_{\theta}(x)] - \mathbb{E}_{\tilde{x} \sim q_{\theta}} [\log p(x)])$$

Reverse KL divergence: fits

Find the optimal parameter, θ^* :

$$\theta^* = \operatorname{argmin}_{\theta} KL(q_{\theta}(x) || p(x))$$

$$= \operatorname{argmin}_{\theta} (\mathbb{E}_{\tilde{x} \sim q_{\theta}} [\log q_{\theta}(x)] - \mathbb{E}_{\tilde{x} \sim q_{\theta}} [\log p(x)])$$

$$= \operatorname{argmax}_{\theta} (-\mathbb{E}_{\tilde{x} \sim q_{\theta}} [\log q_{\theta}(x)] + \mathbb{E}_{\tilde{x} \sim q_{\theta}} [\log p(x)])$$

Reverse KL divergence: fits

Find the optimal parameter, θ^* :

$$\theta^* = \operatorname{argmin}_{\theta} KL(q_{\theta}(x) || p(x))$$

entropy for the
fitted model

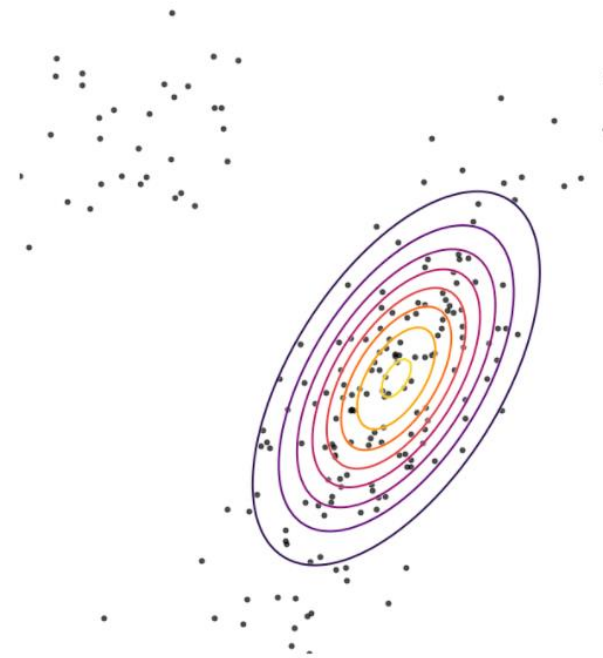
$$\log q_{\theta}(\tilde{x})$$

$$= \operatorname{argmax}_{\theta} (-\mathbb{E}_{\tilde{x} \sim q_{\theta}} [\log q_{\theta}(x)] + \mathbb{E}_{\tilde{x} \sim q_{\theta}} [\log p(x)])$$

relation between
fitted and generated

Reverse KL divergence: fits

- $q_{\theta}(x)$ covers only regions with data
- reasonable in multi-modal data for one solution



Critical: we do not have direct access to $p(x)$.

Jensen-Shannon Divergence



Jensen-Shannon Divergence: idea

- KL divergence is asymmetric

Jensen-Shannon Divergence: idea

- KL divergence is asymmetric

CPD

$$\underline{KL(p||q) + KL(q||p)}$$

Jensen-Shannon Divergence: idea

- KL divergence is asymmetric

- KL can become infinite

$$KL(p||q) + KL(q||p)$$

For KL + Rev KL

$$KL \quad RKL / rKL (P||Q)$$

Jensen-Shannon Divergence: idea

- KL divergence is asymmetric
- KL can become infinite

$$KL(p(x) || \frac{p(x) + q_{\theta}(x)}{2}) + KL(q_{\theta}(x) || \frac{p(x) + q_{\theta}(x)}{2})$$

Jensen-Shannon Divergence: Definition

For $p(x)$ and $q(x)$, two probability distributions,

$$JS(p, q) = \frac{1}{2} \left(KL(p(x) || \frac{p(x) + q_\theta(x)}{2}) + KL(q_\theta(x) || \frac{p(x) + q_\theta(x)}{2}) \right)$$

- symmetric
- nonnegative $0 \leq JS(P, Q) \leq \ln(2)$
- can be transformed to a true distance $\sqrt{JS(p, q)}$

J. Lin Divergence measures based on the Shannon entropy

f -divergences



Definition

- ▶ Let $f: (0; \infty) \rightarrow \mathbb{R}$ be a convex function with $f(1) = 0$.
- ▶ P and Q - two probability distributions on a measurable space $(\mathcal{X}, \mathcal{F})$.
- ▶ p and q - absolutely continuous with respect to a base measure dx defined on $\mathcal{X}$.
- ▶ f -divergence is defined:

$$D_f(P||Q) = \int q(x) f\left(\frac{p(x)}{q(x)}\right) dx$$

- ▶ f is called generator.

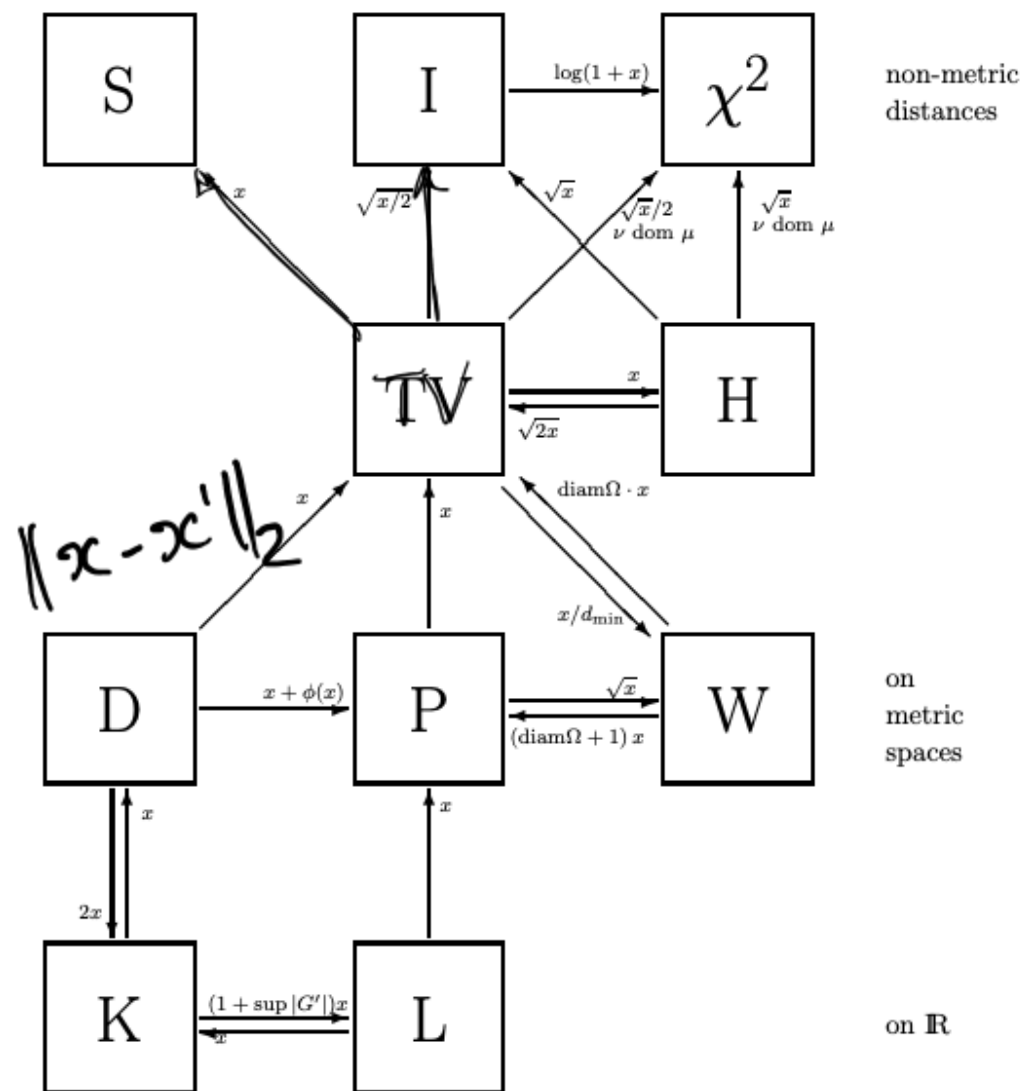
$$D_f(P||Q) = \int q(x) f\left(\frac{p(x)}{q(x)}\right) dx$$

Examples

Name	$D_f(P\ Q)$	Generator $f(u)$
<u>Total variation</u>	$\frac{1}{2} \int p(x) - q(x) \, dx$	$\frac{1}{2} u - 1 $
Kullback-Leibler	$\int p(x) \log \frac{p(x)}{q(x)} \, dx$	$u \log u$
Reverse Kullback-Leibler	$\int q(x) \log \frac{q(x)}{p(x)} \, dx$	$-\log u$
Pearson χ^2	$\int \frac{(q(x) - p(x))^2}{p(x)} \, dx$	$(u - 1)^2$
Neyman χ^2	$\int \frac{(p(x) - q(x))^2}{q(x)} \, dx$	$\frac{(1-u)^2}{u}$
<u>Squared Hellinger</u>	$\int \left(\sqrt{p(x)} - \sqrt{q(x)} \right)^2 \, dx$	$(\sqrt{u} - 1)^2$
Jeffrey	$\int (p(x) - q(x)) \log \left(\frac{p(x)}{q(x)} \right) \, dx$	$(u - 1) \log u$
Jensen-Shannon	$\frac{1}{2} \int p(x) \log \frac{2p(x)}{p(x)+q(x)} + q(x) \log \frac{2q(x)}{p(x)+q(x)} \, dx$	$-(u + 1) \log \frac{1+u}{2} + u \log u$
Jensen-Shannon-weighted	$\int p(x) \pi \log \frac{p(x)}{\pi p(x) + (1-\pi)q(x)} + (1 - \pi)q(x) \log \frac{q(x)}{\pi p(x) + (1-\pi)q(x)} \, dx$	$\pi u \log u - (1 - \pi + \pi u) \log(1 - \pi + \pi u)$

f -divergence inequalities

Abbreviation	Metric
D	Discrepancy
H	Hellinger distance
I	Relative entropy (or Kullback-Leibler divergence)
K	Kolmogorov (or Uniform) metric
L	Lévy metric
P	Prokhorov metric
S	Separation distance
TV	Total variation distance
W	Wasserstein (or Kantorovich) metric
χ^2	χ^2 distance



A. L. Gibbs, F. E. Su On Choosing and Bounding Probability Metrics

Fenchel conjugate

- Each generator has a **Fenchel conjugate** function:

$$f^*(t) = \sup_x (xt - f(x))$$

if f is convex, then $(f^*)^* = f$.

$$f^*(t_1) = \max_{x \in \text{dom}(f)} \{x t_1 - f(x)\}$$

$$x_1 t_1 - f(x_1) \quad f^*(t_1) \quad f^*(t_2) = \max_{x \in \text{dom}(f)} \{x t_2 - f(x)\}$$

$$x_2 t_1 - f(x_2) \quad x_3 t_2 - f(x_3) \quad f^*(t_2)$$

$$x_3 t_1 - f(x_3) \quad x_2 t_2 - f(x_2)$$

$$x_1 t_2 - f(x_1)$$

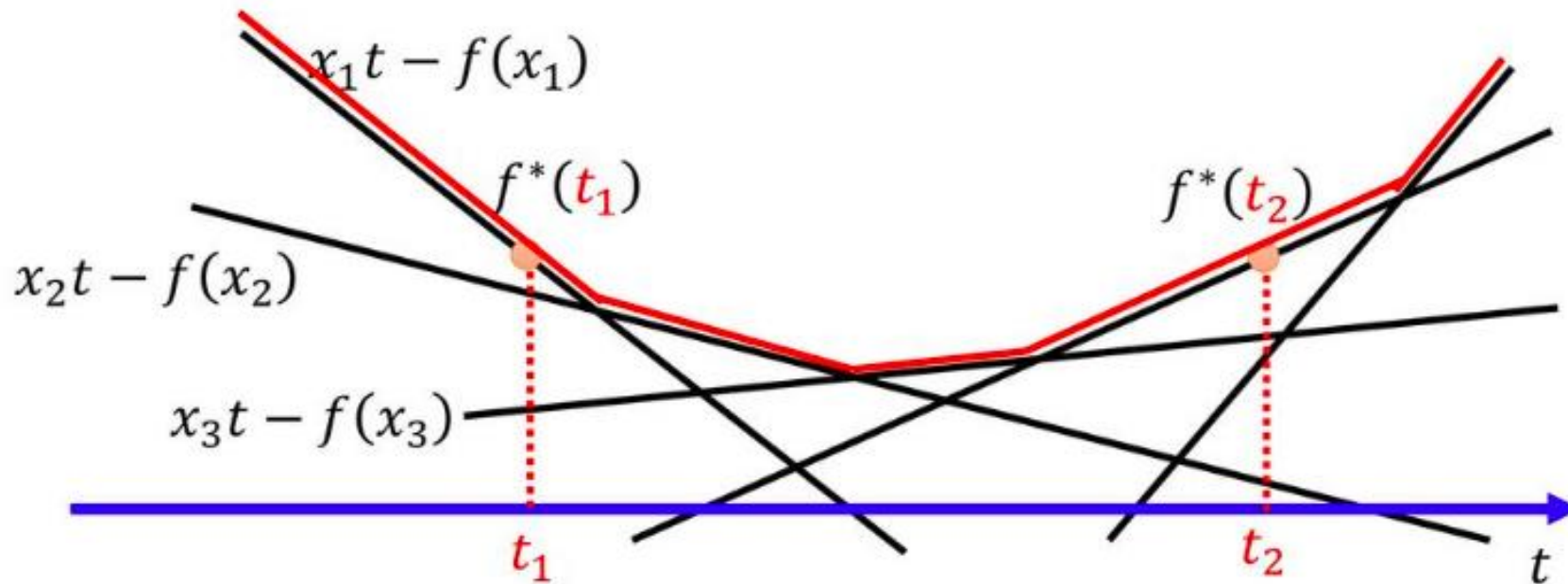


Fenchel conjugate

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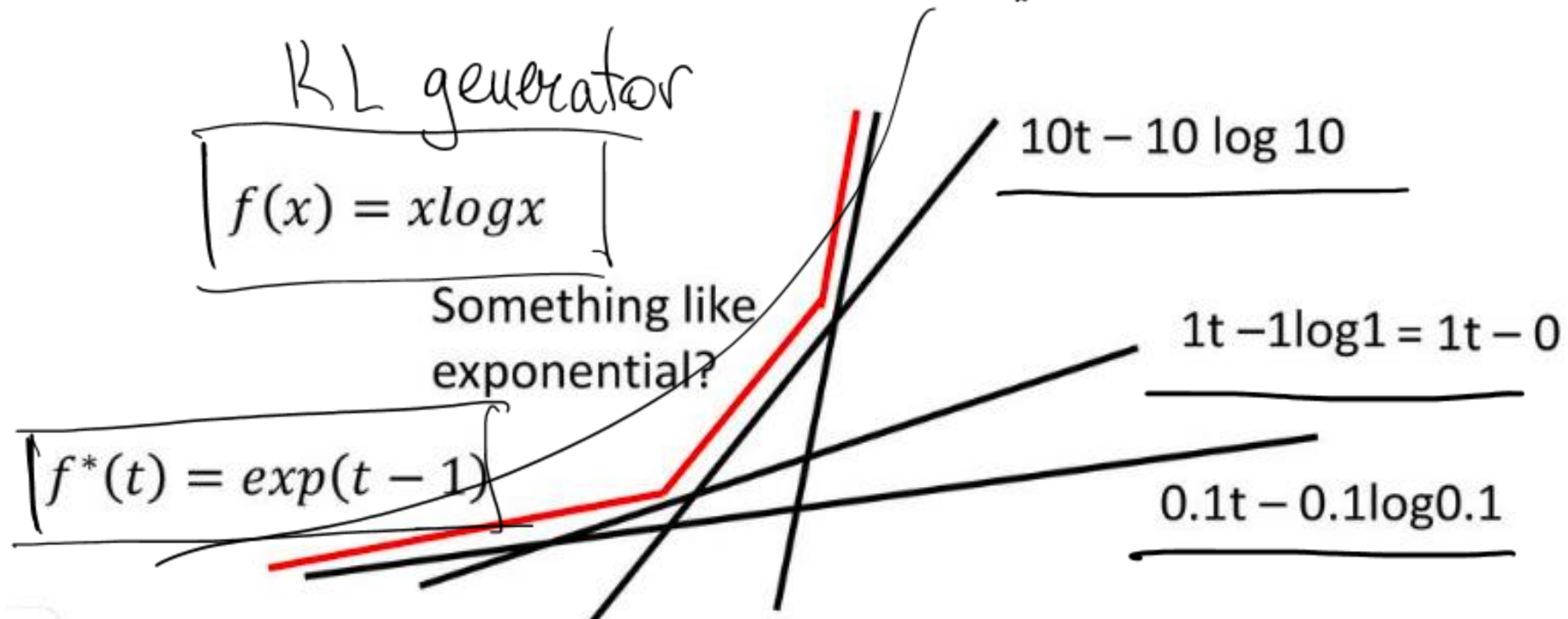


Fenchel conjugate

- Each generator has a **Fenchel conjugate** function:

$$f^*(t) = \sup_x (xt - f(x))$$

if f is convex, then $(f^*)^* = f$.



Fenchel conjugate

$$D_f(P||Q) = \int q(x) f\left(\frac{p(x)}{q(x)}\right) dx$$

- ▶ Each generator has a **Fenchel conjugate** function:

$$f^*(t) = \sup_x (xt - f(x))$$

if f is convex, then $(f^*)^* = f$.

$$f(x) = (f^*)^*(x) = \sup_t (xt - f^*(t))$$

$$\underline{D_f(P||Q) = \int_{\mathcal{X}} q(x) \sup_{t \in \text{dom}_{f^*}} \left\{ t \frac{p(x)}{q(x)} - f^*(t) \right\} dx}$$

$$\underline{\geq \sup_{T \in \mathcal{T}} \left(\int_{\mathcal{X}} p(x) T(x) dx - \int_{\mathcal{X}} q(x) f^*(T(x)) dx \right)}$$

$$\underline{= \sup_{T \in \mathcal{T}} \left(\mathbb{E}_{x \sim P} [T(x)] - \mathbb{E}_{x \sim Q} [f^*(T(x))] \right),}$$

Optimal Variational Function $T(x)$ GAN

- ▶ We can choose $T^*(x)$, an optimal variational function, that makes inequality tightest:

$$T^*(x) = f'\left(\frac{p(x)}{q(x)}\right).$$

$$T^*(x) = f'\left(p(x)/q(x)\right)$$

X. Nguyen et al. Estimating divergence functionals and the likelihood ratio by convex risk minimization

Name	$D_f(P\ Q)$	Generator $f(u)$	$T^*(x)$
Kullback-Leibler	$\int p(x) \log \frac{p(x)}{q(x)} dx$	$u \log u$	$1 + \log \frac{p(x)}{q(x)}$
Reverse KL	$\int q(x) \log \frac{q(x)}{p(x)} dx$	$-\log u$	$-\frac{q(x)}{p(x)}$
Pearson χ^2	$\int \frac{(q(x)-p(x))^2}{p(x)} dx$	$(u-1)^2$	$2\left(\frac{p(x)}{q(x)} - 1\right)$
Squared Hellinger	$\int \left(\sqrt{p(x)} - \sqrt{q(x)}\right)^2 dx$	$(\sqrt{u} - 1)^2$	$\left(\sqrt{\frac{p(x)}{q(x)}} - 1\right) \cdot \sqrt{\frac{q(x)}{p(x)}}$
Jensen-Shannon	$\frac{1}{2} \int p(x) \log \frac{2p(x)}{p(x)+q(x)} + q(x) \log \frac{2q(x)}{p(x)+q(x)} dx$	$-(u+1) \log \frac{1+u}{2} + u \log u$	$\log \frac{2p(x)}{p(x)+q(x)}$

Conclusions

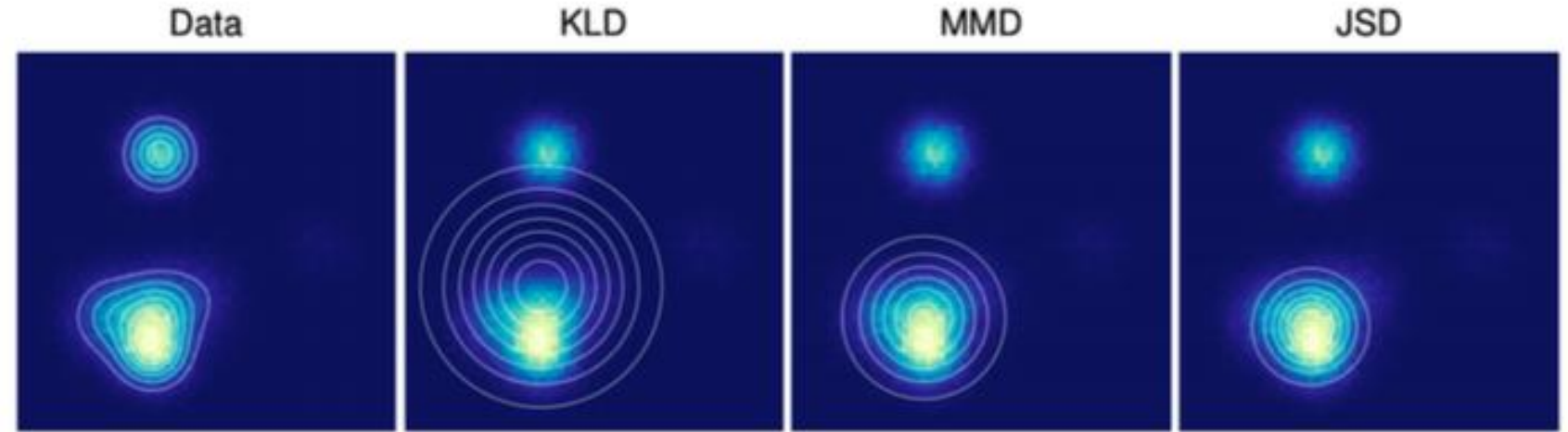


Figure 1: An isotropic Gaussian distribution was fit to data drawn from a mixture of Gaussians by either minimizing Kullback-Leibler divergence (KLD), maximum mean discrepancy (MMD), or Jensen-Shannon divergence (JSD). The different fits demonstrate different tradeoffs made by the three measures of distance between distributions.

Problems:

- ▶ need to use metrics different from optimised;
- ▶ difficulties in case of high dimension problem;
- ▶ not evident choice of a good metric.

<https://arxiv.org/abs/1506.05751>

Conclusions

- f -divergences quantify dissimilarities between probability densities.
 - ▶ Direct optimization of popular divergence requires the knowledge of true function.
- KL & rKL
 - ▶ Special metrics should be developed to quantify quality of generative models (see seminar).
- JS
- ... H, W
- $F_{eu} - conj \rightarrow D_f(P||Q)$